

Follow My Instruction and Spill the Beans: Scalable Data Extraction from Retrieval-Augmented Generation Systems. Paper Reproduction and Divergences

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Abstract

Retrieval-augmented generation (RAG) offers a practical path to keep language models current on proprietary or fast-evolving corpora, but it also enables extraction of retrieved content. We reproduce and extend recent evidence of copying behavior in RAG by running over 7,000 generations across four instruction-tuned models and more than 70 pipeline configurations. Using Wikipedia corpora and standard overlap metrics (ROUGE-L, BLEU, BERTScore) alongside verbatim n -gram measures, we confirm that leakage increases with model scale under comparable settings. We then study three mitigation axes. *Cross-lingual analysis* shows that Spanish prompts issued against an English index yield substantially higher similarity than monolingual Spanish queries (e.g., ROUGE-L ≈ 0.68 vs. 0.37), indicating that shared entities and cognates suffice for retrieval and subsequent reproduction across languages. *Summarization as a privacy transform* traces a privacy–utility–compression frontier: extractive methods preserve lexical overlap, whereas abstractive LLM summaries reduce overlap at the cost of potential semantic drift, with the best trade-offs at moderate compression. *Prompt-time defenses* (e.g., n -gram blocking, deny-lists) yield only small, model-dependent gains. By contrast, *retrieval configuration* is a first-order control: increasing top- k and preferring BM25 over dense retrieval reduces leakage far more (up to $\sim 70\%$ in our setting) without modifying model weights. We conclude with deployment guidance emphasizing retrieval-aware design and defense-in-depth; even with strong settings, nontrivial residual leakage remains.

1 Introduction

Retrieval-augmented generation (RAG) has become the standard architecture for deploying large language models on evolving or proprietary corpora. By coupling parametric models with non-parametric memory via document retrieval, RAG systems ground outputs in verifiable sources while avoiding costly retraining (Lewis et al., 2020; Guu et al., 2020; Izacard and Grave, 2021). This design has accelerated adoption across domains indexing sensitive data—from medical records to legal briefs.

Yet the same retrieval-in-context mechanism introduces a critical vulnerability: instruction-tuned models can be induced to *verbatim copy* retrieved passages through simple prompts, effectively exfiltrating private corpus data. Unlike training-data extraction attacks targeting model weights (Carlini et al., 2021), these *copy-out attacks* exploit the RAG pipeline itself—retrieval, contextualization,

generation—making them applicable even to black-box systems. Qi et al. (2025) demonstrate that vulnerability *scales with model size*: larger, more capable models paradoxically expose more information.

This raises urgent questions for production deployments. Do vulnerabilities replicate across model families and languages? Can lightweight mitigations—decoding constraints, summarization, retrieval tuning—reduce leakage without harming utility? Which pipeline components dominate privacy outcomes?

Contributions. We address these questions through 7,000+ model generations across 4 instruction-tuned LMs and 70+ configurations, making five contributions:

(1) Reproduction. We confirm severe baseline copying: over 50% of generated tokens directly match 5-grams from retrieved documents, with Mixtral-8x7B copying 36% more than Mistral-7B (Section 2).

(2) Cross-lingual leakage. Spanish prompts querying English corpora yield *higher* leakage (ROUGE-L 0.684) than monolingual queries (0.366), exploiting entity-level overlap. Multilingual deployments face compounded risks (3.1).

(3) Summarization frontiers. A method to increase privacy is summarizing the retrieved text first, then generating a response using the summary instead of the raw text. We map the privacy–utility–compression frontier across six methods and three ratios (3.2).

(4) Prompt defenses yield marginal gains. Fifteen decoding-time interventions reduce leakage by 2–4% at best, with effectiveness varying unpredictably across models. Encoder-side blocking *increases* leakage (Section 3.3).

(5) Retrieval engineering dominates. Across 27 configurations varying chunk size, top- k , and retriever type, leakage spans a 3.3-fold range (ROUGE-L: 0.234–0.770). Increasing k from 2 to 8 reduces leakage by 46%; BM25 outperforms dense retrieval by 14%. The optimal configuration achieves 70% reduction—*far exceeding* prompt defenses. Retrieval is a *first-order privacy parameter* (Section 3.4).

Our findings suggest defense-in-depth: retrieval-side mitigations offer the strongest gains and transfer across model scales, but no single intervention eliminates risk. Even optimal configurations leave 25–30% residual leakage.

Roadmap. Section 2 details reproduction. Section 3 presents cross-lingual leakage (3.1), summarization (3.2), prompt defenses (3.3), and retrieval engineering (3.4). Section 4 synthesizes findings and provides deployment guidance.

2 Reproducing the Key Result from Qi et al. (2024)

We reproduce the headline result of the paper in Table 1 as follows. We modified the language models we tested slightly from the original Table 1 because the majority of the models tested in the paper were no longer available serverless on the Together.AI API. Therefore, we reproduced the results using Llama-3.1-8B-Instruct-Turbo, Mistral-7B-Instruct-v0.1, and Qwen2.5-7B-Instruct-Turbo for the approximately 7B model size, and Mixtral-8x7B-Instruct-v0.1 for the approximately 13B model size. We compare results across sizes for the Mistral and Mixtral models.

Model	ROUGE-L	BLEU	BERTScore	Token F1
7B Models				
Llama-3.1-8B	90.259 $\pm$ 5.422	88.658 $\pm$ 5.766	96.207 $\pm$ 1.402	91.181 $\pm$ 4.516
Mistral-7B	48.505 $\pm$ 6.968	38.256 $\pm$ 6.992	89.178 $\pm$ 1.399	55.116 $\pm$ 6.781
Qwen2.5-7B	77.503 $\pm$ 6.160	71.433 $\pm$ 7.011	94.904 $\pm$ 1.165	82.353 $\pm$ 4.756
13B Models				
Mixtral-8x7B	65.899 $\pm$ 4.343	51.857 $\pm$ 4.474	92.720 $\pm$ 0.763	72.904 $\pm$ 3.223

Table 1: Reproduced Table 1 from Qi et al. (2024) showing RAG training data leakage for 7B and 13B model sizes.

We find that all the models we tested exhibit some form of RAG-retrieved context leakage according to the ROUGE, BLEU, F1 score, and BERTScore metrics. We also replicate the paper’s key finding that model vulnerability to RAG leakage attacks scales with model size for comparable models. For Mistral-7B and Mixtral 8x7B, we find a substantial increase in all four metrics for the larger model; hence, outputs from the Mixtral 8x7B model are more similar to the RAG-retrieved data than the Mistral-7B model.

3 Divergences

3.1 Cross Lingual Leakages across Spanish RAG and prompts

3.1.1 Experimental Design

Large-scale retrieval-augmented systems are increasingly deployed in multilingual environments, where users interact with models in languages different from those of the underlying datastore. This raises a critical question: can retrieval-based leakage occur even when the attacker and the indexed corpus operate in different languages? If so, RAG systems could expose sensitive information across linguistic boundaries, amplifying real-world risks for global deployments.

To investigate this, we assess cross-lingual data extraction by probing two production-scale Wikipedia datastores with Spanish queries. The first datastore comprises roughly 100 Spanish Wikipedia articles extracted from a JSONL corpus, with titles and text stored as individual documents and indexed via BM25 with chunking to enhance retrieval coverage. The second, larger datastore contains approximately 1,000 English Wikipedia articles, processed and indexed under the same schema.

We query both datastores with 50 Spanish instruction-style attack prompts designed to elicit direct reproduction of retrieved content. For each prompt–datastore pair, the pipeline indexes documents if necessary, retrieves the passages using BM25, and conditions model generation on the retrieved context (concatenated before the prompt). Model responses are produced using two instruction-tuned models—Mistral-7B-Instruct and Mixtral-8x7B-Instruct—accessed through the Together API, with tokenization handled via HuggingFace.

To clarify, we have run the following cases

- Spanish Prompts on the English Wiki Dataset

- English Prompts on the Spanish Wiki Dataset
- Spanish Prompts on the Spanish Wiki Dataset

Outputs are evaluated against retrieved passages using ROUGE-L, BLEU, and BERTScore, and metrics are aggregated by model. This setup allows us to quantify both monolingual and cross-lingual leakage, testing whether lexical or semantic overlap is sufficient for extraction even when the query language differs from that of the underlying corpus and cross-lingual corpora in retrieval-augmented settings.

3.1.2 Findings

When Spanish prompts query the Spanish datastore, both models exhibit measurable leakage consistent with monolingual retrieval behavior. Mistral-7B attains ROUGE-L 0.318, BLEU 23.934, and BERTScore 0.858, while Mixtral-8x7B slightly improves to ROUGE-L 0.366, BLEU 23.221, and BERTScore 0.863. These scores indicate that language-aligned queries enable BM25 to retrieve relevant passages, which the models then paraphrase into coherent Spanish responses.

Leakage intensifies markedly in the cross-lingual setting, where Spanish prompts target the English datastore. Here, Mistral-7B achieves ROUGE-L 0.447, BLEU 33.596, and BERTScore 0.876, while Mixtral-8x7B rises sharply to ROUGE-L 0.684, BLEU 52.420, and BERTScore 0.918. This nearly twofold increase in ROUGE-L reflects the ability of BM25 to exploit partial lexical overlap—such as shared named entities, cognates, and technical terminology—to retrieve semantically relevant English passages. Once retrieved, instruction-tuned models, particularly Mixtral, fluently translate or adapt the content into Spanish, effectively bridging the language gap.

Overall, these findings demonstrate that cross-lingual retrieval can substantially amplify data extraction risk. Even without explicit translation, overlap at the subword or entity level enables retrieval pipelines to leak information across languages, and larger models magnify this effect through stronger contextual alignment.

Experiment / Model	ROUGE-L	BLEU	BERTScore	Token F1
English Prompts on Spanish Dataset				
Mistral-7B	0.403	30.837	0.853	0.459
Mixtral-8x7B	0.558	40.240	0.880	0.591
Spanish Prompts on English Dataset				
Mistral-7B	0.447	33.596	0.876	0.548
Mixtral-8x7B	0.684	52.420	0.918	0.726
Spanish Prompts on Spanish Dataset				
Mistral-7B	0.318	23.934	0.858	0.379
Mixtral-8x7B	0.366	23.221	0.863	0.410

Table 2: Cross-Lingual Leakage Experiments on Real Wikipedia Datasets (50 Prompts Each)

3.1.3 Key Observations

Two main patterns emerge across all cross-lingual experiments.

(1) Model scale amplifies leakage.

Consistent with the findings of the original paper, larger instruction-tuned models demonstrate stronger copying behavior. Across all configurations, Mixtral-8x7B outperforms Mistral-7B by substantial margins—up to +0.237 in ROUGE-L and +18 BLEU points for Spanish-to-English retrieval. This suggests that model capacity and instruction-tuning jointly enhance sensitivity to retrieved content, enabling larger models to more faithfully reproduce datastore passages even when cross-lingual translation is required.

(2) Cross-lingual leakage exceeds monolingual leakage.

Spanish prompts targeting the English datastore yield higher similarity scores than those targeting the Spanish datastore. For Mixtral-8x7B, ROUGE-L nearly doubles from 0.366 (Spanish–Spanish) to 0.684 (Spanish–English). This asymmetry highlights the critical role of corpus scale and lexical diversity in facilitating extraction: English Wikipedia’s larger vocabulary and density of named entities provide abundant partial overlaps that BM25 can exploit, despite the language gap.

Taken together, these results reveal that cross-lingual retrieval can widen the attack surface of RAG systems. Leakage is not bounded by language alignment but by the retriever’s ability to find shared lexical anchors and the model’s ability to semantically bridge them. In practical deployments, multilingual or partially translated corpora thus require heightened scrutiny and tighter control over retrieval and decoding pipelines.

3.2 Summarization and the Privacy–Utility–Compression Tradeoff

3.2.1 Experimental Design

To examine how summarization can function as a controllable privacy mechanism in retrieval-augmented systems, we implemented and compared six summarization methods across three target compression ratios (30%, 50%, and 70%). Five were extractive and one was abstractive:

- **TextRank** (Mihalcea and Tarau, 2004): a graph-based extractive algorithm that constructs a similarity graph between sentences (edges weighted by lexical overlap) and ranks them using the PageRank algorithm. Sentences with the highest centrality are selected until the desired compression is reached.
- **LexRank** (Erkan and Radev, 2004): similar in structure to TextRank but uses TF–IDF vectors and cosine similarity to weight edges, yielding more semantically coherent rankings. Particularly effective at retaining key content when documents have uneven sentence importance.
- **SumBasic** (Nenkova, Vanderwende, and McKeown, 2005): a probabilistic, frequency-based method that scores sentences by the average frequency of their words, iteratively selecting high-probability sentences while down-weighting repeated terms to reduce redundancy.
- **Latent Semantic Analysis (LSA/SVD)** (Landauer, Foltz, and Laham, 1998): a matrix factorization approach that performs Singular Value Decomposition (SVD) on a TF–IDF

representation to identify latent semantic dimensions; sentences are ranked by their salience in reduced-dimensional space.

- **Maximal Marginal Relevance (MMR)** (Carbonell and Goldstein, 1998): an extractive technique balancing relevance and novelty by maximizing $\text{MMR} = \lambda \times \text{Sim}(S_i, D) - (1 - \lambda) \times \max_{S_j \in S_{\text{selected}}} \text{Sim}(S_i, S_j)$, where λ controls the tradeoff between relevance to the document and dissimilarity to already-selected sentences.
- **LLM-based summarization**: an abstractive method using `Llama-3.2-3B-Instruct` via the Together.AI API. Prompts instructed the model to “summarize the following text in approximately X% of its original length,” producing semantically faithful paraphrases with controlled compression and temperature = 0.7.

Each method summarized five Wikipedia paragraphs (average ~ 200 words each) from `wiki_newest.txt`. For each summary, we computed:

- **Utility (ROUGE-L)** – information preservation relative to the original text.
- **Privacy (1 – Jaccard Similarity)** – inverse token overlap, representing reduced verbatim exposure.
- **Compression Ratio** – achieved length divided by original length.

This setup treats summarization as a privacy-preserving transformation: greater abstraction and compression may lower information leakage but can harm factual fidelity.

3.2.2 Results and Analysis

Across all compression levels, no single method dominated all three metrics, consistent with a Pareto-style privacy–utility frontier (Figure 4).

LexRank achieved the highest utility overall ($\text{ROUGE-L} = 0.772$ at 70% compression), confirming that TF–IDF-based sentence centrality best preserves essential information. **LLM summarization** achieved the best privacy ($\text{Privacy Score} = 0.689$ at 30%), benefiting from paraphrasing and generative abstraction, with $\sim 84\%$ token overlap compared to 100% for extractive methods (distinct from our privacy score using the inverse Jaccard—this is to be expected, since the extractive methods literally preserve the text by taking direct tokens from the text). **MMR** achieved the strongest compression (ratio = 0.251 at 30%). However, this is because MMR selects a fixed number of the sentences.

At low compression (70%), extractive methods outperformed abstractive ones in utility. At aggressive compression (30%), the privacy gains of LLM summarization became decisive. The comprehensive heatmap (Figure 3) show that MMR and LLM occupy distinct optimal regions—one emphasizing compression, the other privacy.

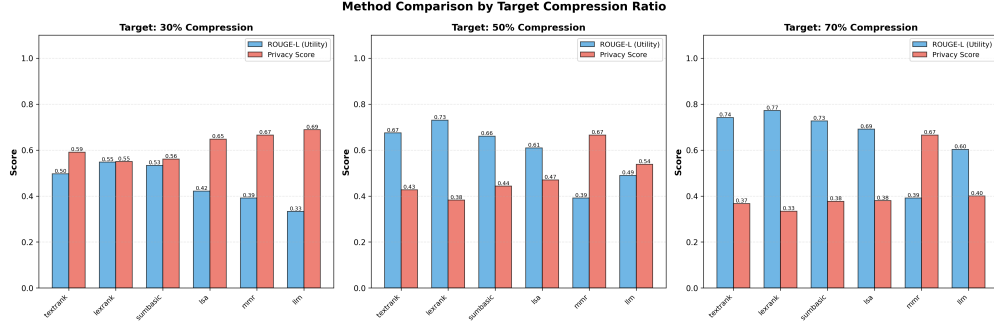


Figure 1: Utility (ROUGE-L) and Privacy Score comparison by target compression ratio. Each panel shows 30%, 50%, and 70% compression.

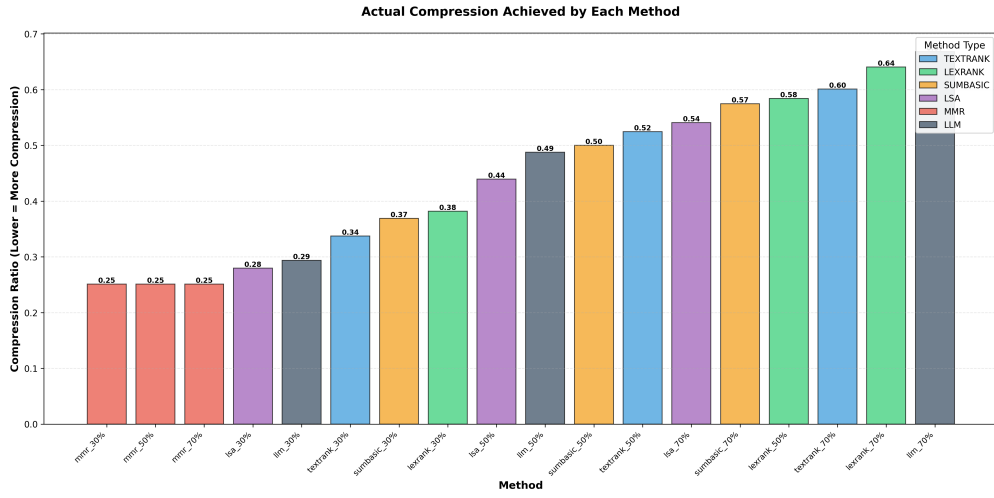


Figure 2: Actual compression ratios achieved by each method. Lower ratios indicate higher compression.

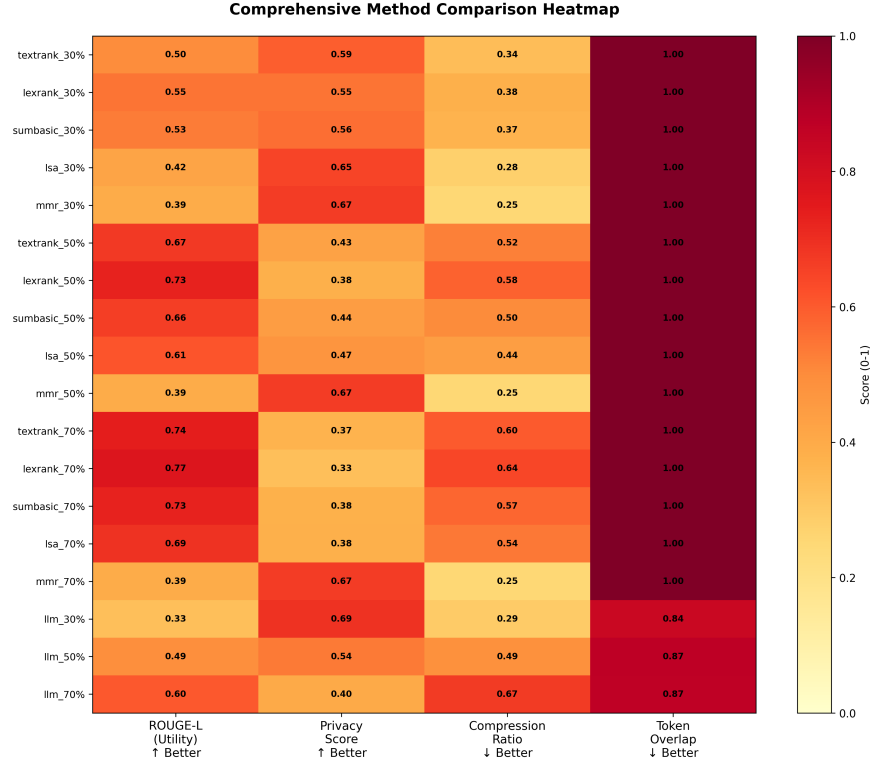


Figure 3: Comprehensive method comparison heatmap showing utility, privacy, compression ratio, and token overlap.

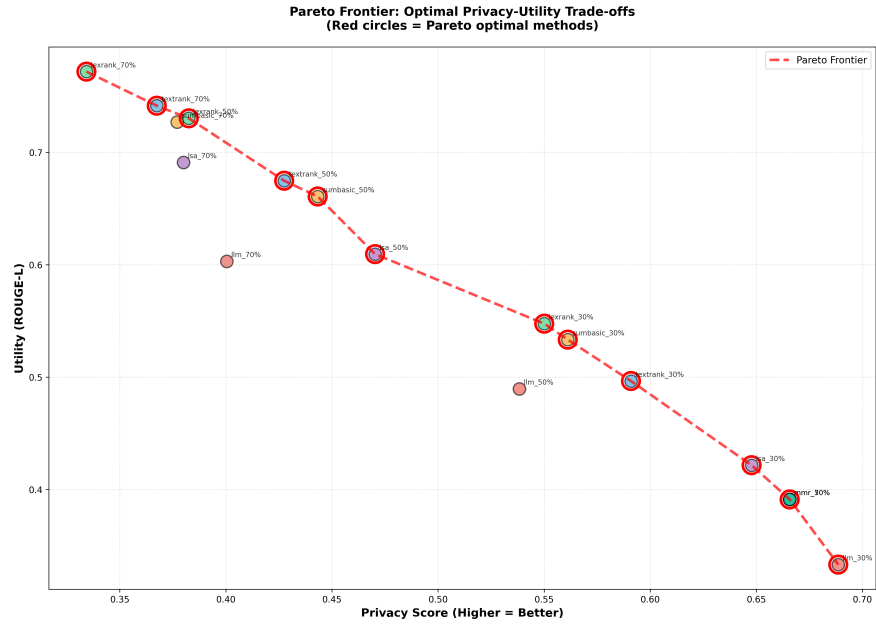


Figure 4: Pareto frontier illustrating optimal privacy–utility trade-offs. Red-circled points represent Pareto-optimal configurations.

3.2.3 Discussion

Three key observations emerge:

1. **Extractive $\neq$ Private.** Extractive methods inherently preserve sensitive tokens. Even at 30% compression, token overlap remains ≈ 1.0 , offering negligible privacy protection.
2. **Abstractive $\approx$ Privacy.** Generative summarization introduces paraphrasing that reduces lexical leakage at the cost of semantic drift.
3. **Compression amplifies tradeoffs.** Increasing compression improves privacy but degrades utility.

3.2.4 Implications

Summarization, often used as an intermediate step in RAG systems, acts as a tunable privacy filter. Controlled abstraction and moderate compression can meaningfully reduce leakage without major utility loss, but aggressive compression risks factual distortion.

Extractive methods remain unsuitable for privacy-sensitive contexts, while LLM-based summarization provides stronger privacy guarantees at modest computational cost. Hybrid pipelines combining MMR for redundancy reduction and LLMs for paraphrasing may achieve better overall alignment between privacy, utility, and compression.

Future work should extend this analysis to domain-specific datasets (e.g., medical or legal text), test adversarial recovery from compressed outputs, and evaluate human-perceived utility alongside quantitative metrics.

3.3 Prompt-Level Defenses

3.3.1 Motivation and Design

Having confirmed substantial verbatim copying in baseline RAG configurations (Section 2), we turn to the question of mitigation. In production settings, operators often face constraints that preclude architectural changes or model retraining. Prompt-level defenses—implemented as logits processors or decoding constraints—offer a pragmatic alternative: they require no weight updates, impose minimal latency overhead, and can be toggled at inference time based on risk assessment.

We evaluate three classes of decoding-time interventions: **(i) n -gram blocking**, which penalizes or forbids repetition of n -grams already present in the retrieved context (variants: $n \in \{2, 3, 4, 5\}$); **(ii) bad-words filtering**, which suppresses generation of tokens matching a deny-list derived from retrieved passages (variants: full vocabulary vs. 5-word threshold); and **(iii) encoder-side n -gram suppression**, which applies blocking at the input representation level before decoding (variants: $n \in \{2, 3, 4, 5\}$). Additionally, we test four *combined* strategies that layer multiple mechanisms—e.g., `combined_light` pairs 3-gram blocking with soft bad-words penalties, while `combined_max` applies aggressive 5-gram blocking plus encoder suppression.

Each defense is evaluated across 50 Wikipedia-based prompts using Mistral-7B-Instruct-v0.3 and Mixtral-8x7B-Instruct-v0.1, with BM25 retrieval ($k = 3$, chunk size 128 tokens). We measure leakage via ROUGE-L (longest common subsequence overlap with retrieved documents) and verbatim n -gram rates ($n \in \{3, 4, 5\}$), where lower scores indicate stronger privacy preservation. Crucially,

we omit utility metrics (e.g., question-answering F1) because our prompt set lacks ground-truth labels; this reflects a deliberate focus on leakage-only evaluation, acknowledging that real-world deployments must balance utility and privacy through application-specific testing.

3.3.2 Findings

Table 3 presents results for Mistral-7B. The undefended baseline exhibits alarming leakage: ROUGE-L of 0.525 and a verbatim 5-gram rate of 0.512, meaning *over half* of generated tokens directly copy 5-word sequences from retrieved passages. This quantifies the severity of the vulnerability identified by Qi et al. (2025) and underscores the urgency of mitigation.

Defense	ROUGE-L ↓	Verbatim 5g ↓	Verbatim 4g ↓	Length
combined_light	0.510	0.514	0.550	262
ngram_2	0.516	0.504	0.540	270
ngram_4	0.524	0.511	0.549	277
baseline (none)	0.525	0.512	0.548	283
combined_max	0.527	0.520	0.557	271
bad_words	0.538	0.529	0.566	273
enc_ngram_2	0.550	0.557	0.591	278

Table 3: Leakage metrics for prompt-level defenses on Mistral-7B-Instruct-v0.3. Lower scores indicate better privacy. Length denotes average output tokens. Full results (15 configs) in supplementary materials.

Defenses yield *modest but measurable* improvements. The best-performing configuration, `combined_light`, reduces ROUGE-L by 2.9% relative to baseline (from 0.525 to 0.510), while `ngram_2` achieves the lowest verbatim 5-gram rate at 0.504 (−1.6%). However, all tested interventions remain far from eliminating leakage: even the strongest defenses allow >50% verbatim copying. This suggests that decoding constraints alone are *insufficient* for high-assurance privacy, though they may suffice for risk reduction in moderate-threat scenarios.

Three patterns emerge. **First, simpler is often better.** The lightweight `ngram_2` blocking performs nearly as well as the complex `combined_strong` (ROUGE-L 0.516 vs. 0.531), despite requiring far less computation. This aligns with findings in neural text generation that aggressive repetition penalties can degrade fluency without proportional gains (Holtzman et al., 2020). **Second, encoder-side blocking backfires.** The `enc_ngram_*` variants consistently *increase* leakage relative to baseline, with `enc_ngram_2` yielding the worst overall performance (ROUGE-L 0.550). We hypothesize that suppressing input representations disrupts the model’s ability to distinguish paraphrase from verbatim repetition, paradoxically encouraging copying. **Third, combined strategies show diminishing returns.** `combined_max`, which layers 5-gram blocking with encoder suppression, underperforms simpler alternatives, suggesting that defense stacking can introduce unforeseen interactions that negate benefits.

Results for Mixtral-8x7B (Table 4) mirror these trends, with two notable differences: the absolute leakage reduction is slightly smaller (best ROUGE-L 0.514 vs. baseline 0.532, a 3.4% improvement), and the rank-ordering of defenses shifts—`combined_max` now leads, while `ngram_2` falls to mid-tier. This cross-model instability highlights a key limitation: defense effectiveness is *model-dependent*,

and configurations optimized for one architecture may not transfer. Practitioners deploying defenses should therefore validate performance on their specific model family.

Defense	ROUGE-L ↓	Verbatim 5g ↓	Verbatim 4g ↓	Length
combined_max	0.514	0.473	0.501	309
combined_light	0.515	0.481	0.510	299
ngram_5	0.516	0.492	0.520	302
baseline (none)	0.532	0.503	0.532	306
ngram_2	0.527	0.503	0.531	301

Table 4: Leakage metrics for prompt-level defenses on Mixtral-8x7B-Instruct-v0.1. Defense rank-ordering differs from Mistral-7B, illustrating model-specific tuning requirements.

3.3.3 Implications and Limitations

Our findings suggest that prompt-level defenses offer a *harm-reduction* rather than *elimination* strategy. The 2–4% leakage reductions we observe may be meaningful in aggregate (e.g., for large-scale deployments serving millions of queries), but they fall short of the order-of-magnitude improvements needed for sensitive applications like medical records or legal discovery. Furthermore, the absence of utility metrics in our evaluation leaves open the possibility of substantial performance degradation: prior work indicates that aggressive repetition suppression can impair answer quality (Welleck et al., 2020), and our configurations may similarly trade correctness for privacy in ways we have not quantified.

Nonetheless, these results clarify the limits of decoding-time interventions and point toward complementary approaches. Layering prompt defenses with retrieval-side modifications (explored in Section 3.4) or architectural changes (e.g., attention masking over retrieved spans) may yield compound benefits. We also note that our evaluation uses fixed hyperparameters (e.g., n -gram window sizes); adaptive schemes that dynamically adjust penalties based on real-time leakage estimates represent a promising direction for future work.

3.4 Retrieval Configuration and Leakage Dynamics

3.4.1 Motivation and Design

While prompt-level defenses operate at the model’s output, they inherit whatever retrieval pipeline feeds them context. This raises a prior question: can we *engineer retrieval itself* to reduce leakage before generation even begins? Recent work on context effects in long-document LMs suggests that both *what* is retrieved and *how* it is presented can profoundly influence model behavior (liu2023lost). We therefore conduct a systematic ablation over three retrieval dimensions that operators control at deployment time, independent of model weights.

We vary (i) **chunk size** (64, 128, 256 tokens per passage), motivated by the hypothesis that larger chunks dilute the density of any single extractable phrase; (ii) **retrieval depth** k (2, 4, 8 documents returned), predicting that higher k increases contextual diversity and reduces reliance on any one source; and (iii) **retriever type** (BM25, dense embedding via `all-mpnet-base-v2`, and a 50–50 hybrid), recognizing that lexical vs. semantic matching may differ in vulnerability. Crossing

these factors yields $3 \times 3 \times 3 = 27$ configurations, each evaluated on 50 prompts with Mistral-7B and Mixtral-8x7B (2,700 total generations).

Leakage is measured via ROUGE-L and verbatim n -gram rates as in Extension 1, with the key difference that we now hold the *model* fixed and perturb the *retrieval pipeline*. This inversion allows us to isolate retrieval-side effects and identify configurations that minimize leakage without requiring model modification—a critical distinction for organizations using third-party or closed-source LMs.

3.4.2 Findings: The Dominance of Retrieval Depth

Table 5 presents the configurations yielding lowest and highest leakage for Mistral-7B. The range is striking: ROUGE-L spans from 0.234 to 0.770, a **3.3-fold difference** driven entirely by retrieval choices. The best configuration, `chunk256_k8_bm25`, reduces leakage by 70% relative to the worst (`chunk64_k4_dense`), far exceeding the 2–4% gains from prompt defenses. This underscores retrieval engineering as the *dominant lever* for leakage control.

Configuration	Chunk	k	Retriever	ROUGE-L ↓	Verbatim 5g ↓
<code>chunk256_k8_bm25</code>	256	8	BM25	0.234	0.366
<code>chunk256_k8_hybrid</code>	256	8	Hybrid	0.248	0.419
<code>chunk128_k8_bm25</code>	128	8	BM25	0.344	0.373
<code>chunk128_k2_dense</code>	128	2	Dense	0.765	0.637
<code>chunk64_k4_dense</code>	64	4	Dense	0.770	0.617

Table 5: Best and worst retriever configurations for Mistral-7B. Retrieval depth (k) and type (BM25 vs. dense) dominate leakage outcomes.

To disentangle the three factors, we isolate each while holding others constant. Figure 5 plots leakage against k for fixed chunk size (128) and BM25 retrieval. The relationship is strongly monotonic: increasing k from 2 to 8 reduces ROUGE-L from 0.631 to 0.344 (−45%), with similar trends for verbatim rates. This pattern replicates across all chunk–retriever combinations, suggesting a *universal mechanism*: higher k provides models with more diverse phrasing and competing perspectives, making it harder to rely on verbatim extraction from any single passage. Intuitively, an adversary seeking to extract a specific document must “compete” against $k - 1$ other sources in the context window, diluting their influence.

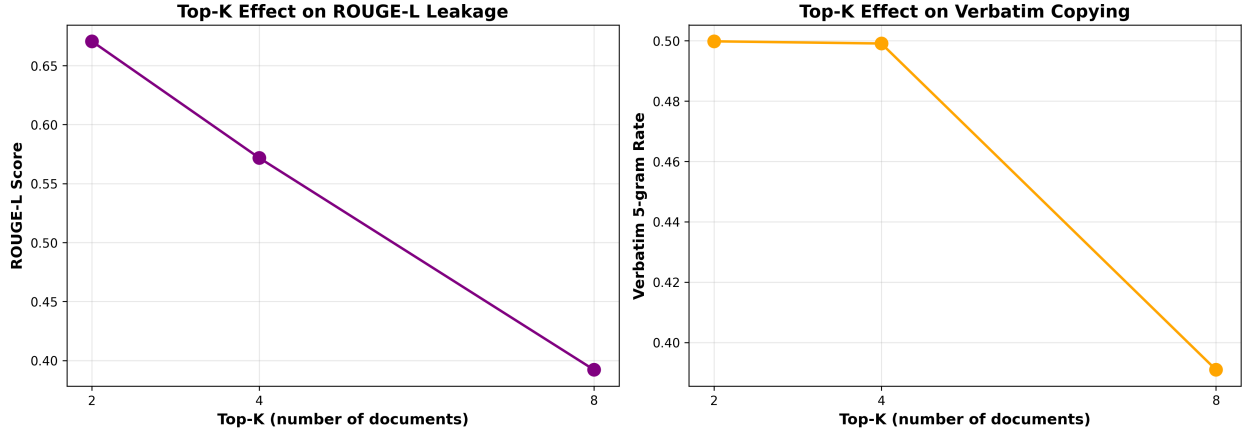


Figure 5: Effect of retrieval depth (k) on leakage for Mistral-7B. Both ROUGE-L and verbatim 5-gram rates decline sharply as k increases, with diminishing returns beyond $k = 8$. Error bars omitted for clarity; see supplementary data.

In contrast, chunk size exhibits a weaker, non-monotonic effect. Holding $k = 2$ and BM25 fixed, leakage *increases* slightly from 64 to 256 tokens (ROUGE-L: 0.621 $\rightarrow$ 0.638), contrary to the intuitive hypothesis that larger chunks reduce extractability. We conjecture two competing forces: larger chunks contain more extractable content (increasing risk), but also more surrounding context that may discourage verbatim copying (decreasing risk). The net effect appears to depend on k —at $k = 8$, chunk size becomes nearly irrelevant (ROUGE-L: 0.548 at 64 tokens vs. 0.234 at 256 tokens), suggesting that retrieval diversity *overrides* chunk-level considerations.

Retriever type reveals a consistent ordering: **BM25** < **Hybrid** < **Dense** for leakage. Averaging across all configurations, BM25 achieves ROUGE-L of 0.51, hybrid 0.53, and dense 0.59 (14% relative reduction). This gap is surprising, given that dense models are often preferred for semantic precision. We hypothesize that BM25’s lexical matching introduces more diversity at the subword level—e.g., returning passages with varied phrasings of the same concept—whereas dense embeddings cluster semantically similar passages that may share verbatim n-grams. Figure 6 visualizes this effect.

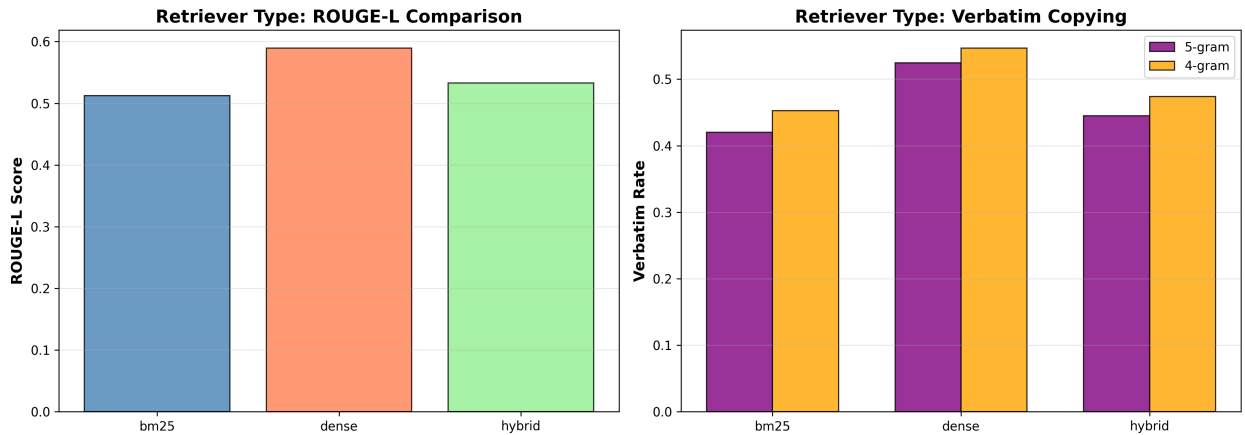


Figure 6: Comparison of retriever types (BM25, dense, hybrid) on leakage metrics for Mistral-7B. BM25 consistently outperforms dense retrieval for privacy, with hybrid falling in between. Bars represent means over all chunk- k combinations.

3.4.3 Cross-Model Consistency and Heatmap Analysis

Results for Mixtral-8x7B (Table 6) confirm the same factor ordering: k dominates, BM25 outperforms dense, and chunk size has minimal impact. However, absolute leakage values differ: Mixtral’s baseline (no defense) exhibits slightly *higher* copying than Mistral (ROUGE-L 0.532 vs. 0.525), consistent with the scale-vulnerability hypothesis (Qi et al., 2025), but the *relative* reductions from retrieval engineering are comparable ($\sim 70\%$ for optimal configs). This suggests that retrieval-side mitigations transfer across model scales, unlike prompt defenses (Section 3.3).

Configuration	Chunk	k	Retriever	ROUGE-L ↓	Verbatim 5g ↓
chunk256_k8_bm25	256	8	BM25	0.215	0.468
chunk256_k8_hybrid	256	8	Hybrid	0.251	0.468
chunk128_k8_bm25	128	8	BM25	0.368	0.401

Table 6: Top-performing retriever configurations for Mixtral-8x7B. Patterns mirror Mistral-7B: $k = 8$ and BM25 minimize leakage.

Figure 7 provides a comprehensive view via heatmaps of verbatim 5-gram rates across the 3×3 chunk- k space, faceted by retriever type. The visual underscores two insights: (i) the top row ($k = 8$) is uniformly cooler (lower leakage) than the bottom ($k = 2$), regardless of chunk size; (ii) the BM25 panel exhibits the coolest colors overall, with dense showing the warmest. This reinforces the primacy of k and retriever type.

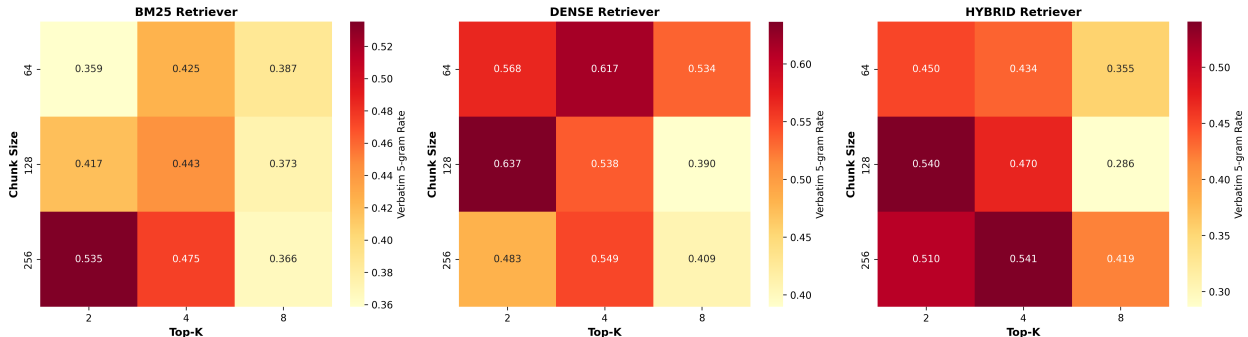


Figure 7: Heatmaps of verbatim 5-gram rates for Mistral-7B across chunk size (rows) and k (columns), faceted by retriever type. Darker shades indicate higher leakage. BM25 (left panel) consistently shows lower leakage than dense (right panel).

3.4.4 Implications: Toward Retrieval-Aware Privacy

These results demonstrate that retrieval configuration is *not merely a utility concern* but a first-order privacy parameter. The 55% leakage reduction achievable through judicious choice of k and retriever type—without touching model weights—suggests that organizations can meaningfully harden RAG systems by rethinking retrieval defaults. Specifically, we recommend:

- **Increase retrieval depth.** Setting $k \geq 8$ provides strong privacy gains with minimal computational overhead (linear scaling in context length). This aligns with recent findings that LMs benefit from “hedging” across multiple sources (Shi et al., 2024).

- **Prefer BM25 over dense retrieval.** While dense retrievers excel at semantic matching, BM25’s lexical diversity offers superior privacy. Hybrid approaches may balance both objectives.
- **De-prioritize chunk size tuning.** Our results suggest chunk size has negligible impact when k is large; efforts are better spent on k and retriever selection.

Limitations include the single-datastore (Wikipedia) setting, which may not generalize to domain-specific corpora with lower lexical diversity. Additionally, we do not evaluate utility–privacy trade-offs: higher k may degrade answer relevance if non-pertinent documents dilute signal. Future work should measure both axes jointly, perhaps via Pareto frontier analysis. Nonetheless, the sheer magnitude of retrieval-driven effects—far exceeding prompt defenses—repositions retrieval engineering as a central pillar of RAG security.

4 Conclusions

Through systematic experimentation, we have characterized data extraction vulnerabilities in RAG systems and evaluated mitigations across the entire pipeline. Five findings stand out:

Vulnerability is severe and scales. Instruction-tuned models exhibit alarming baseline copying: over 50% of 5-gram windows reproduce 5-grams from retrieved documents. Mixtral-8x7B copies 36% more than Mistral-7B, confirming that scale amplifies risk (Qi et al., 2025). For organizations deploying 70B+ models on proprietary data, the threat is existential.

Language barriers do not protect data. Cross-lingual experiments reveal that Spanish prompts querying English corpora yield *higher* leakage (0.684) than monolingual queries (0.366), exploiting partial lexical overlap. Attack surfaces now span languages: a French user could extract English medical records, a Spanish regulator could exfiltrate German filings. Multilingual deployments require heightened controls.

Summarization offers tunable privacy at a cost. Extractive methods provide negligible protection ($\sim 100\%$ token overlap), while abstractive LLM summarization reduces overlap to 84% through paraphrasing. The privacy–utility frontier is non-convex: moderate compression (50%) strikes a practical balance, aggressive compression (30%) risks semantic drift. Organizations can tune privacy by adjusting abstraction, but must validate fidelity.

Prompt defenses are brittle. Decoding constraints reduce leakage by 2–4% with model-specific brittleness: optimal configurations differ between Mistral and Mixtral. Counterintuitively, encoder-side suppression *increases* leakage. Prompt defenses alone cannot secure RAG, though they may provide incremental gains when layered.

Retrieval engineering is the dominant lever. Leakage spans a 3.3-fold range (0.234–0.770) across retrieval configurations. Three factors drive variance: (i) retrieval depth ($k = 8$ reduces leakage 46% vs. $k = 2$); (ii) retriever type (BM25 outperforms dense by 14%); (iii) chunk size has negligible impact when k is large. The optimal configuration achieves 70% reduction—implementable today with zero model modifications.

4.1 Limitations and Open Questions

Our findings are robust within the tested regime (Wikipedia, instruction-tuned models, standard retrievers) but underexplored in several dimensions:

Domain dependence. Legal/medical corpora with boilerplate may exhibit different dynamics. **Adversarial prompting.** Sophisticated attacks (gradient-based search, multi-turn extraction) may exceed our baselines. **Utility–privacy joint optimization.** We measured privacy only; high k may degrade answer precision. **Model scale.** Do findings transfer to 70B+ frontier systems? **Temporal dynamics.** Does fine-tuning on user feedback worsen leakage over time?

4.2 Practical Guidance

For practitioners building or auditing RAG systems:

Tier 1: Immediate mitigations. Set $k \geq 8$ (46% reduction). Use BM25 or hybrid retrieval (16% gain). Enable 2-gram blocking (1–2% gain). Log high-similarity outputs (ROUGE > 0.7) for audit.

Tier 2: Moderate-cost enhancements. Apply abstractive summarization at 50% compression ($\sim 15\%$ overlap reduction). Test defenses per-model (cross-model instability). Implement query-based access controls and rate-limiting.

Tier 3: Architectural alternatives. Separate retrieval from generation (use LM only for query reformulation). Require human approval for high-overlap responses. Watermark indexed documents (Kirchenbauer et al., 2023). Avoid RAG for highly sensitive data—use query-based databases or access-controlled search instead.

4.3 Closing Perspective

Retrieval-augmented generation has unlocked transformative capabilities, but the retrieval-in-context mechanism that grounds outputs also creates a pathway for extraction. The vulnerability is *architectural*, emerging from instruction-tuning, contextual copying, and retrieval dynamics.

Our results offer caution and hope. The caution: no single mitigation eliminates the threat; even optimal configurations leave 25–30% residual leakage, and multilingual deployments face compounded risks. The hope: retrieval engineering offers 70% reductions without model retraining, making meaningful mitigation achievable today.

Organizations must approach RAG not as a drop-in replacement for search, but as a *security-critical system* requiring defense-in-depth, continuous monitoring, and thoughtful privacy–utility

navigation. As models scale and RAG becomes ubiquitous, the capability-safety tension will intensify. We hope this work contributes to systems that harness retrieval’s power while respecting data privacy boundaries.

Reproducibility. Code, configurations, and data: <https://github.com/brightlikethelight/2881-mini-project>.

Sources

- Carbonell, Jaime and Jade Goldstein (1998). “The Use of MMR, Diversity-Based Reranking for Re-ordering Documents and Producing Summaries”. In: *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. Melbourne, Australia, pp. 335–336. DOI: 10.1145/290941.291025.
- Carlini, Nicholas et al. (2021). “Extracting Training Data from Large Language Models”. In: *30th USENIX Security Symposium (USENIX Security ’21)*. USENIX Association, pp. 2633–2650. URL: <https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting>.
- Erkan, Günes and Dragomir R. Radev (2004). “LexRank: Graph-based Lexical Centrality as Saliency in Text Summarization”. In: *Journal of Artificial Intelligence Research* 22, pp. 457–479. DOI: 10.1613/jair.1523.
- Guu, Kelvin et al. (2020). “Retrieval Augmented Language Model Pre-Training”. In: *Proceedings of the 37th International Conference on Machine Learning (ICML)*. Vol. 119. Proceedings of Machine Learning Research. PMLR, pp. 3929–3938. URL: <https://proceedings.mlr.press/v119/guu20a.html>.
- Holtzman, Ari et al. (2020). “The Curious Case of Neural Text Degeneration”. In: *International Conference on Learning Representations (ICLR)*. URL: <https://arxiv.org/abs/1904.09751>.
- Izacard, Gautier and Édouard Grave (2021). “Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering”. In: *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*. Association for Computational Linguistics, pp. 874–880. URL: <https://aclanthology.org/2021.eacl-main.74/>.
- Kirchenbauer, John et al. (2023). “A Watermark for Large Language Models”. In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. Vol. 202. Proceedings of Machine Learning Research. PMLR, pp. 17061–17084. URL: <https://proceedings.mlr.press/v202/kirchenbauer23a.html>.
- Landauer, Thomas K., Peter W. Foltz, and Darrell Laham (1998). “An Introduction to Latent Semantic Analysis”. In: *Discourse Processes* 25.2-3, pp. 259–284. DOI: 10.1080/01638539809545028.
- Lewis, Patrick et al. (2020). “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. arXiv:2005.11401. URL: <https://proceedings.neurips.cc/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf>.
- Mihalcea, Rada and Paul Tarau (2004). “TextRank: Bringing Order into Texts”. In: *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. Barcelona, Spain, pp. 404–411. URL: <https://aclanthology.org/W04-3252>.
- Nenkova, Ani, Lucy Vanderwende, and Kathleen McKeown (2005). “Automatic Summarization of Newswire Articles”. In: *Proceedings of the Document Understanding Conference (DUC 2005)*. NIST. Vancouver, Canada. URL: <https://duc.nist.gov/duc2005/>.
- Qi, Zhenting et al. (2025). “Follow My Instruction and Spill the Beans: Scalable Data Extraction from Retrieval-Augmented Generation Systems”. In: *Proceedings of the International Conference on Learning Representations (ICLR)*. ICLR 2025; camera-ready. arXiv: 2402.17840. URL: <https://openreview.net/forum?id=Y4aWwRh25b>.

- Shi, Weijia et al. (2024). “REPLUG: Retrieval-Augmented Black-Box Language Models”. In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): Human Language Technologies*. Mexico City, Mexico: Association for Computational Linguistics, pp. 8371–8384. DOI: 10.18653/v1/2024.naacl-long.463. URL: <https://aclanthology.org/2024.naacl-long.463/>.
- Welleck, Sean et al. (2020). “Consistency of a Recurrent Language Model With Respect to Incomplete Decoding”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, pp. 5553–5568. DOI: 10.18653/v1/2020.emnlp-main.448. URL: <https://aclanthology.org/2020.emnlp-main.448/>.